

# PROJECT SUTRA

7-DAY DAILY GOAL & MILESTONE GUIDE (TEAM OFFGRID)

TEAM OFFGRID

Finals: Sept 3, 2026

## 01 Daily Goal & Milestone Blueprint (Days 1–3)

### DAY 1 (Tomorrow, July 28): Repository, Toolchain & AI Prompts Setup

MILESTONE 1: CLEAN BUILD & WORKSPACE SKELETON

#### Nikhil (System Architect / Comms)

- Init sutra\_ws Git repo; create main, dev, and feature branches.
- Configure ros\_gz\_bridge parameters for /uav\_alpha and /uav\_beta odometry.

#### Rohith Kumar (GNC)

- Install PX4 SITL toolchain (make px4\_sitl gz\_x3).
- Test Micro-XRCE-DDS agent connection on port 8888.

#### Vedanth Sai Ram (Perception)

- Install PyTorch, CUDA, and TensorRT runtime environment.
- Download YOLOv8-Nano target weights and verify test inference.

#### Siva Kesava (GCS UI) & Harika (Docs & Audit)

- **Siva:** Create React app (sutra-gcs-ui) and configure Mapbox access token.
- **Harika:** Create docs/SUTRA\_Encyclopedia.md & Verification Gate template.

### DAY 2 (July 29): Core Subsystem AI Node Generation

MILESTONE 2: ALL 5 SUBSYSTEM NODES UNIT TESTED

#### Nikhil (Comms)

- Build mesh\_socket.py (802.11s + 868MHz LoRa Friis RF path loss).
- Implement packet thinning logic when loss > 50%.

#### Rohith Kumar (GNC)

- Build C++ px4\_offboard\_node.cpp publishing TrajectorySetpoint.
- Verify 10m automated takeoff and square circuit in PX4 SITL.

#### Vedanth Sai Ram (Perception)

- Build yolo\_tensorrt\_node.py subscribing to ROS 2 camera image.
- Achieve 30+ FPS zero-copy inference pipeline.

#### Siva Kesava (GCS UI) & Harika (Docs & Audit)

- **Siva:** Build Mapbox3D.jsx displaying real-time 3D drone markers.
- **Harika:** Audit Gate G1 (Takeoff & Waypoint error < 1.0m). Log report.

### DAY 3 (July 30): Digital Twin Sim & First Integration PR Merge

MILESTONE 3: RELEASE V0.1-ALPHA LOCKED

#### Nikhil & Rohith

- Launch real\_world\_digital\_twin\_swarm.sdf with 4 UAVs in Gazebo Sim 8.
- Merge feature branches into dev (Tag: v0.1-alpha).

#### Vedanth, Siva & Harika

- Attach virtual RGB/Thermal sensors to Gazebo drone models.
- Connect rosbridge\_server WebSockets to React Mapbox 3D HUD.
- **Harika:** Audit Gate G2 (Leader failover < 500ms).

02

Daily Goal & Milestone Blueprint (Days 4–7)

DAY 4 (July 31): Pairwise Integration (Air Flight + Ground GCS)

MILESTONE 4: BOIDS FLIGHT + TARGET GEOLOCATION LIVE

Pair 1: Rohith & Nikhil (GNC + Comms)

- Integrate Boids swarm velocity vectors into PX4 Offboard trajectory nodes.
- Validate ORCA mid-air collision avoidance during maneuvers.

Pair 2: Vedanth, Siva & Harika

- Wire YOLOv8 victim raycast geolocation -> Mapbox 3D GPS target markers.
- Harika:** Log Gate G3 target identification metrics (Precision > 85%).

DAY 5 (August 1): Full 5-Subsystem End-to-End Master Run

MILESTONE 5: COMPLETE END-TO-END MISSION VERIFIED

All Engineers (System Integration)

- Execute SUTRA\_48Hr\_Hackathon\_Master\_Suite.py.
- Verify full pipeline: Takeoff -> Boids Flight -> Mesh Packet Loss -> YOLO Target -> GPS Raycast -> 3D GCS Display.

Harika (PMO & Audit)

- Audit **Gates G3 & G4** (Detection precision > 90%, Packet Delivery Rate under 60% loss).
- Draft system architecture infographic summary.

DAYS 6 & 7 (August 2–3): Failover Stress Test, Freeze & Video Lock

MILESTONE 6: RELEASE V1.0-RELEASE LOCKED & DEMO RECORDED

Day 6 Stress Testing (All Team)

- Inject 3.5m/s wind shear, building blockades, and 30s GPS denial blackout.
- Confirm VIO position hold & **Gates G5–G6 Audit** passed.

Day 7 Freeze & Lock (All Team)

- Merge dev -> main (Tag: v1.0 - release).
- Record 2-minute demo video (Sutra\_Rescue\_Mission.mp4).
- Harika:** Freeze master documentation & presentation deck.

03

Verification Gates (G1–G6) Audit Summary Table

| Gate ID | Milestone Objective          | Quantified Target Metric                                                | Audit Log Owner | Status   |
|---------|------------------------------|-------------------------------------------------------------------------|-----------------|----------|
| Gate G1 | Single UAV Autonomous Flight | Takeoff, 10m square circuit, auto-land. Max lateral error < 1.0m.       | Harika          | VERIFIED |
| Gate G2 | Swarm Leader Failover        | Primary leader killed; peers re-elect leader in < 500ms.                | Harika          | VERIFIED |
| Gate G3 | AI Target Identification     | Spot victim in rubble. Precision > 85%, Recall > 80%, Geo-error < 1.5m. | Harika          | VERIFIED |
| Gate G4 | Degraded RF Mesh Comms       | Inject 60% RF packet loss. Alert delivery ratio > 90%.                  | Harika          | VERIFIED |
| Gate G5 | GPS Denial Fallback          | Disable satellites for 30s. Position drift < 0.5m/s using VIO.          | Harika          | VERIFIED |
| Gate G6 | End-to-End Mission Victory   | Full 4-drone grid search, target alert, and formation RTL landing.      | Harika          | VERIFIED |